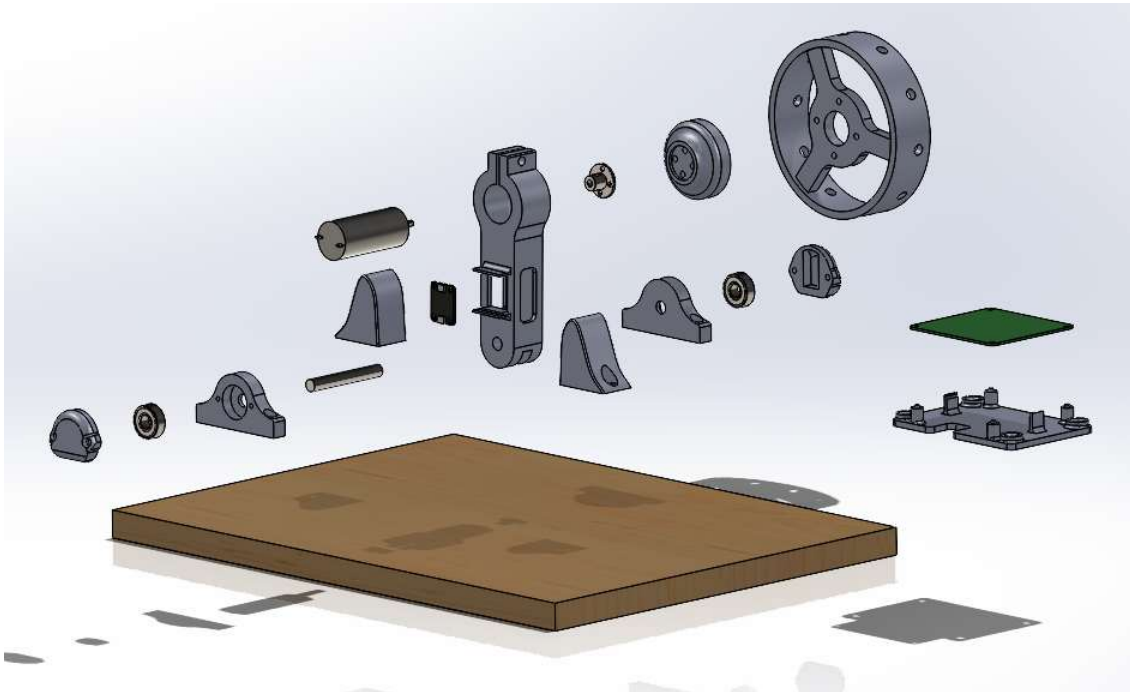


Inverted Pendulum Reaction Wheel

Instructions:

Pendulum Setup



(There are fasteners and wires missing from this image. This includes: 12x M3 screws, connectors for the motor and imu, 10x timber screws for base mounting and a bolt and nut set for the motor mount)

Tools needed: Phillips screwdriver, tiny phillips screwdriver, M3 allen key

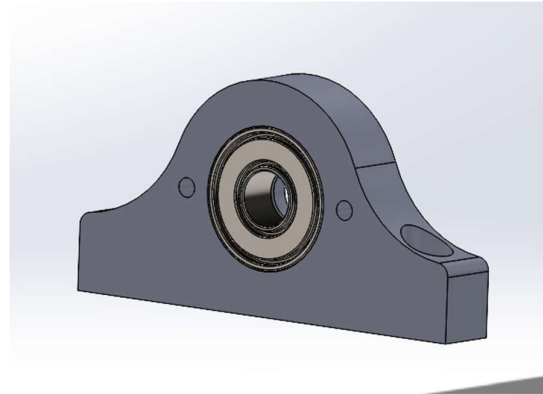
30 September 2026

1. Hinge and Bearing

Insert one bearing into each of the two bearing housings on the base.

The bearings should be seated flush inside the hinge.

Do not force the bearings into position at an angle.

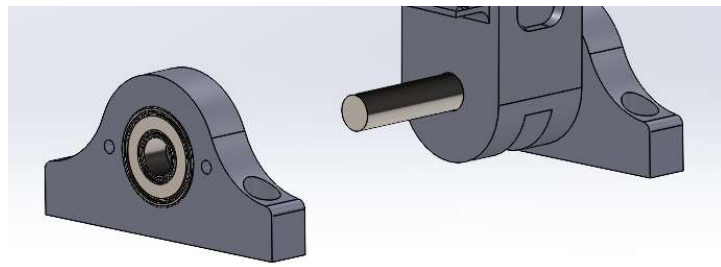


2. Shaft

Slide the shaft through the first bearing, pendulum arm and into the second bearing.

Ensure the shaft passes freely through both bearings.

The arm should rotate smoothly without significant resistance.



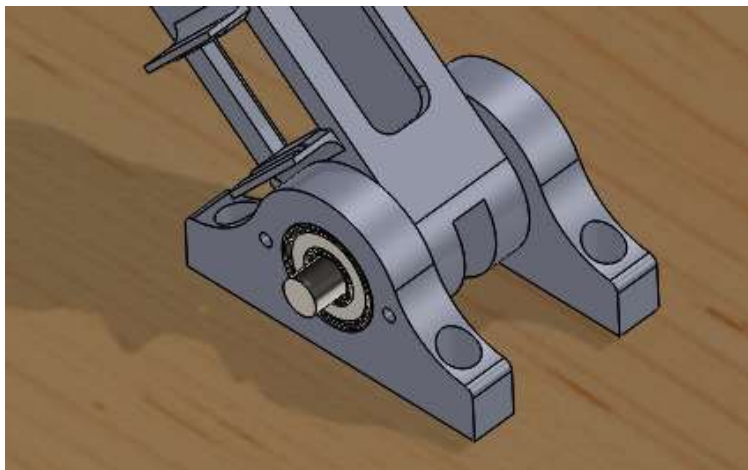
3. Hinge mount

Position the two hinges on the base board according to the specified hole locations. The rough and exposed wood edge of the board should be facing away and at the back.

Secure each hinge to the board using four timber screws, tightening them evenly to ensure the hinges remain straight and aligned.

Check that both hinges are parallel to each other before fully tighten.

It is very important to prevent any friction from keeping the pendulum from moving freely.

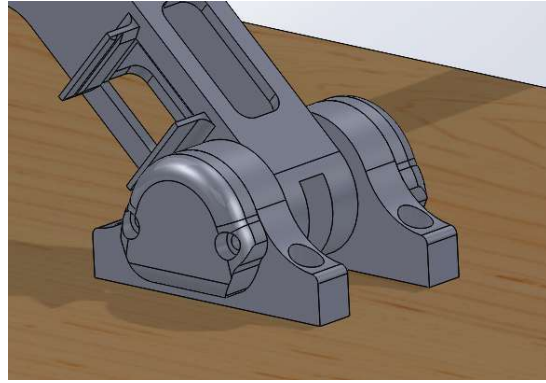


4. Shaft Covers

Place the bearing and shaft covers over the bearings and shaft ends on both sides of the base.

Secure the covers using M3 screws, ensuring they sit flush against the hinges.

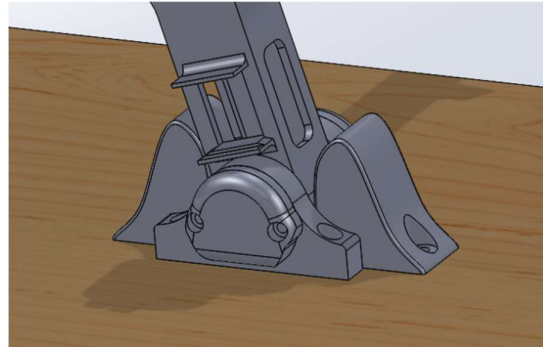
Check that the covers do not interfere with the shaft or restrict its rotation.



5. Arm Supports

Position the two arm supports on the base board according to the specified hole locations. Use the closest holes to the arm for now. The other holes can be used if the starting angle needs to be changed.

Secure each arm support to the board using two timber screws.

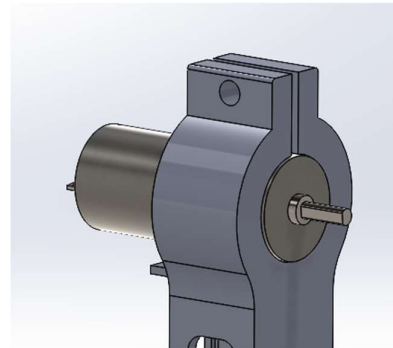


6. Motor

Place the motor into the motor mount integrated into the pendulum arm.

Ensure the motor is fully seated in the mounting feature and that the motor face is flush with the arm as shown.

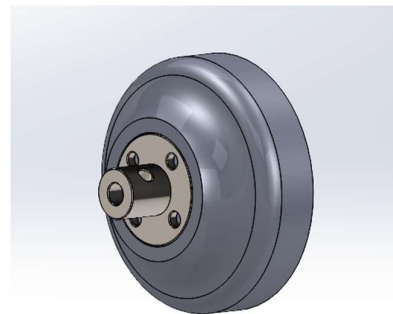
Secure the motor using the nut and bolt.



7. Wheel Adapter

Attach the wheel adapter to the clamp hub using 4x M3 screws.

Tighten the four screws progressively in a cross pattern to keep the wheel evenly secured to the hub.



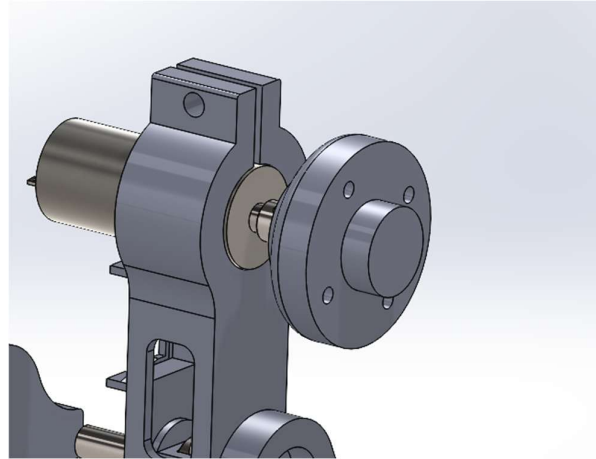
8. Clamp Hub

Attach the clamp hub to the motor shaft.

Ensure the hub is fully seated on the motor's D-shaped shaft and is aligned.

Secure the hub using the preset M3 screws. You can align one of these screws with the D in the shaft.

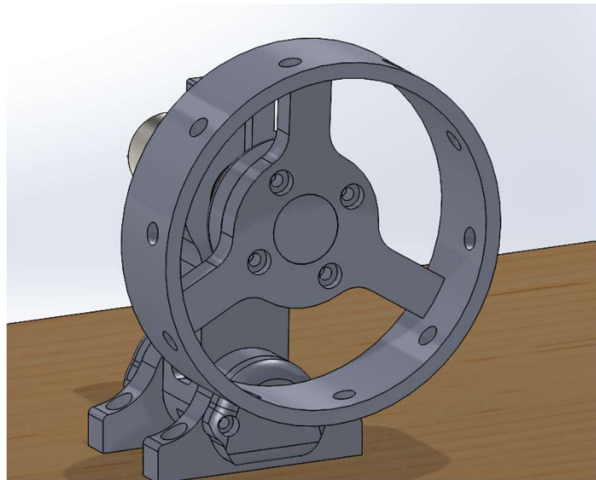
Check that the hub does not wobble excessively when rotated manually.



9. Reaction Wheel

Attach the reaction wheel to the wheel adapter.

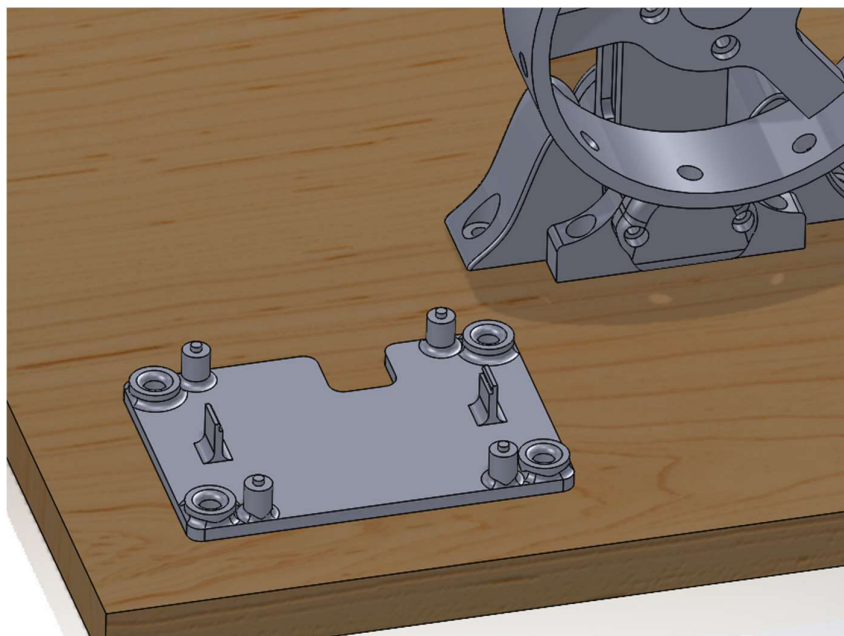
Tighten the screws progressively in a cross pattern with 4x M3 screws.



10. PCB Holder

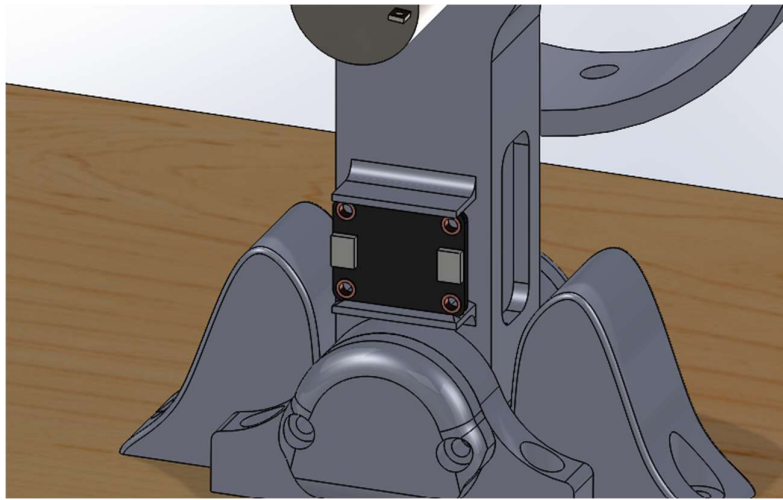
Position the PCB holder on the base board according to the specified hole locations.

Secure the holder to the board using four timber screws.



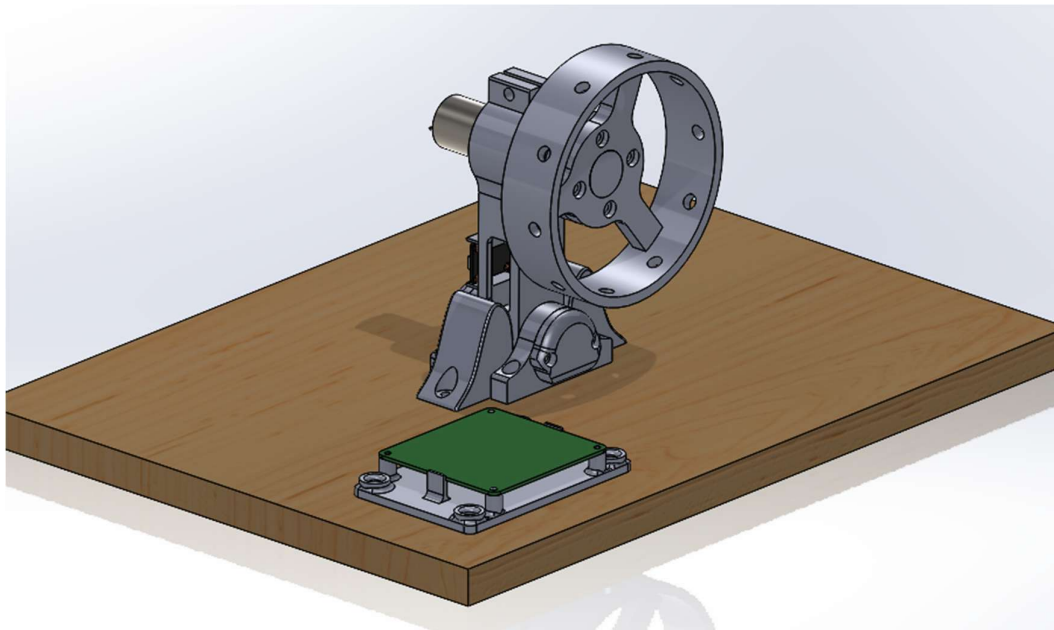
11. IMU

Clip the IMU gently into the integrated slot on the back of the pendulum arm.



Final Assembly

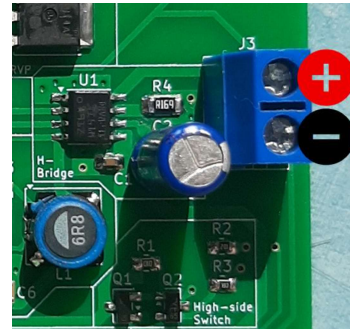
(edit for pcb holder location possibly)



Board Setup

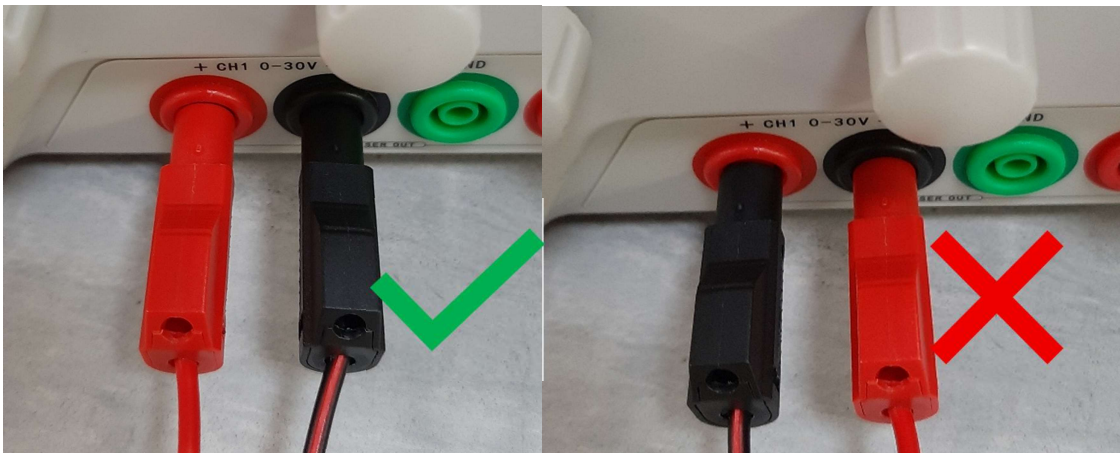
Wiring

Before you power on the board, you must ensure the motor and IMU are connected to the blue screw terminals located on either side of the board. The IMU has 4 connections located on the left of the board while the motor has two. The IMU also has 2 optional connections labelled RST and INT that aren't necessary. Following the silkscreen printing on the PCB and IMU to connect them together properly. The red wire on the motor should be connected to the top screw terminal and the black on the bottom. This can be seen in the reference photo to the right. Once this is done you are ready to power on the board.



Power

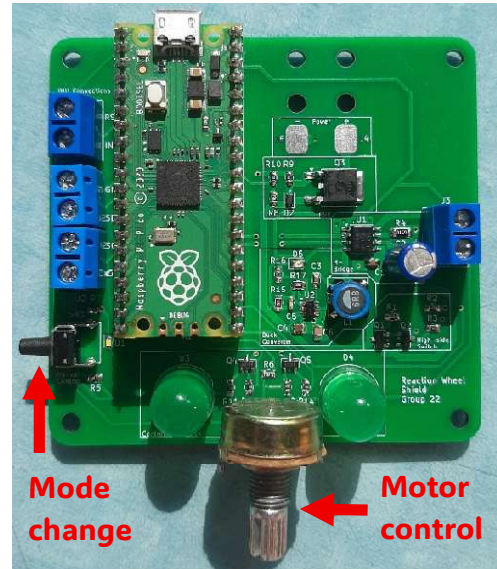
For power the board can use a bench power supply which you can connect with the banana cables that the board has. It is important that the board is connected the right way otherwise the board will not function. Make sure you connect the red cable to the positive terminal of your power supply, and the black cable to the negative end. See the photos below:



Make sure the supply is set to 12V and be sure to keep the current limit at around 3 Amps. Once the system is on there will be an orange light that will turn on to indicate that the Raspberry Pi Pico has power. If the board has an input voltage of 12V, then the motor will begin spinning.

Board Controls

Once the board has power, there are a couple controls. First is the “mode change” switch which will toggle between manual mode and controls mode. This is located on the bottom left of the board as seen in the reference image. Controls mode is the default mode where the board will use a control loop to keep the pendulum upright. Manual mode turns off this loop and requires the user to use the dial (located at the bottom of the PCB) to keep the pendulum upright. The farther the dial is turned in one direction, the faster the motor spins.



Recalibration

If the Raspberry Pi Pico needs to be replaced or recalibrated with the code, it can be connected to the board with a USB plug. Please note that the board cannot be powered with this alone, but will not be damaged when this occurs. Please be sure keep the PCB unplugged from the power supply when doing this as to not damage the system. Once the code has been uploaded to the Pico, make sure to unplug the USB cable before operating.

Safety systems:

The PCB has a couple safety features in case of improper use of the system. It has reverse polarity protection in case the input voltage is connected the way around. It also has some overvoltage protection up to 30V. Any higher and the PCB will be damaged.

The PCB holder will be mounted onto the board to display the PCB safely. The PCB clips onto the holder through two clips in addition to four divots to keep the PCB in position. This also ensures that the PCB will not be accidentally bumped or yanked off the board.

Arm holders are screwed into the board to keep the pendulum arm from falling. This will also keep the wheel safe and prevent the wheel from rubbing against the board. The code also includes a safe guard on top of this, limiting the angle that the pendulum will try to correct. Once the pendulum exceeds this angle, the motor will stop spinning.